

DSA 210: Introduction to Data Science – Final Project Report

Project Title: Predicting Movie Success: Does Financial Power Guarantee Audience Satisfaction?

Student Name: Zeynep Canlı

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1. Motivation

The motion picture industry is a high-stakes economic ecosystem where production budgets for major studio films regularly exceed hundreds of millions of dollars. A prevailing, multi-billion-dollar assumption among industry executives is that financial power and heavy investment directly correlate with the final quality, emotional reach, and success of a cinematic release.

The core motivation of this data science project is to empirically test, evaluate, and challenge this corporate assumption using real-world public data. Specifically, this study addresses a fundamental question: Does a massive budget guarantee audience happiness, or is audience appreciation driven by intrinsic, qualitative factors that capital cannot buy? By mapping the relationship between objective financial inputs and subjective audience sentiment outputs, this project aims to uncover whether structural metadata can successfully model and predict human emotional resonance.

2. Data Source & Enrichment Pipeline

In strict accordance with the core project guidelines regarding the use of multiple public datasets, this project explicitly enriched its analytical architecture by cleanly merging two distinct, comprehensive repositories sourced from Kaggle:

- **TMDB 5000 Movies Dataset:** Utilized as the foundational repository for structural and financial vectors, including budget, revenue, raw genre metadata arrays, and historical release dates.

- **Rotten Tomatoes Movies Dataset:** Utilized as the secondary enrichment layer to secure verified critical and user reception metrics, specifically the continuous `audience_rating` attribute.

Data Preprocessing & Cleaning Steps:

- **JSON Parsing & Extraction:** Because the TMDB dataset stores genre metadata inside structurally serialized JSON strings, a custom parsing routine was implemented to programmatically isolate and extract each film's primary genre label, transforming dirty nested structures into a clean categorical feature.
- **Dataset Integration (Enrichment via Inner Join):** To create a unified data matrix, an explicit relational inner join operation was executed across both datasets, matching records strictly on normalized text keys (`original_title` from TMDB and `movie_title` from Rotten Tomatoes). This process successfully combined financial history and audience emotion into a single dataframe.
- **Anomalous Data Truncation & Filtering:** To prevent highly skewed conclusions caused by incomplete historical logs or placeholder zero-entries, films with unrealistic financial metrics were strictly dropped using a combined non-zero baseline constraint: `(budget > 10,000) and (revenue > 10,000)`.
- **Feature Engineering (Return on Investment - ROI):** A new efficiency metric, ROI, was mathematically engineered to capture financial performance relative to initial cost risk: $ROI = (revenue - budget) / budget$
- **Target Variable Standardization:** The raw, multi-scale user score attribute was isolated, cast to a high-precision floating-point number, and rescaled to a standard continuous interval between 0.0 and 1.0, establishing the final project target variable: `happiness_score`.

3. Data Analysis & Methodology

The data science exploration and modeling pipeline was structured and executed across three carefully designed chronological phases:

Phase 1: Exploratory Data Analysis & Non-Parametric Testing

Initial distribution checks via multi-genre box plots demonstrated that the continuous spread of `happiness_score` across distinct genres (Action, Adventure, Drama, Comedy, Crime, Horror) severely violated parametric normality assumptions due to localized skewness, heavy tails, and asymmetric user behavior. Consequently, non-parametric statistical methods were mandatory to ensure mathematical validity:

- **Spearman's Rank Correlation Coefficient (Rho):** Calculated as a robust non-parametric measure to evaluate the global strength and direction of the monotonic relationship between a film's financial budget and its final audience score.
- **Kruskal-Wallis Test:** Deployed as a robust non-parametric analysis of variance. Movies were binned into four logical, ordinal financial tiers based on empirical percentiles (Low, Medium, High, Very High) to evaluate whether statistical distributions of audience happiness significantly differ when treated as macro-groups.

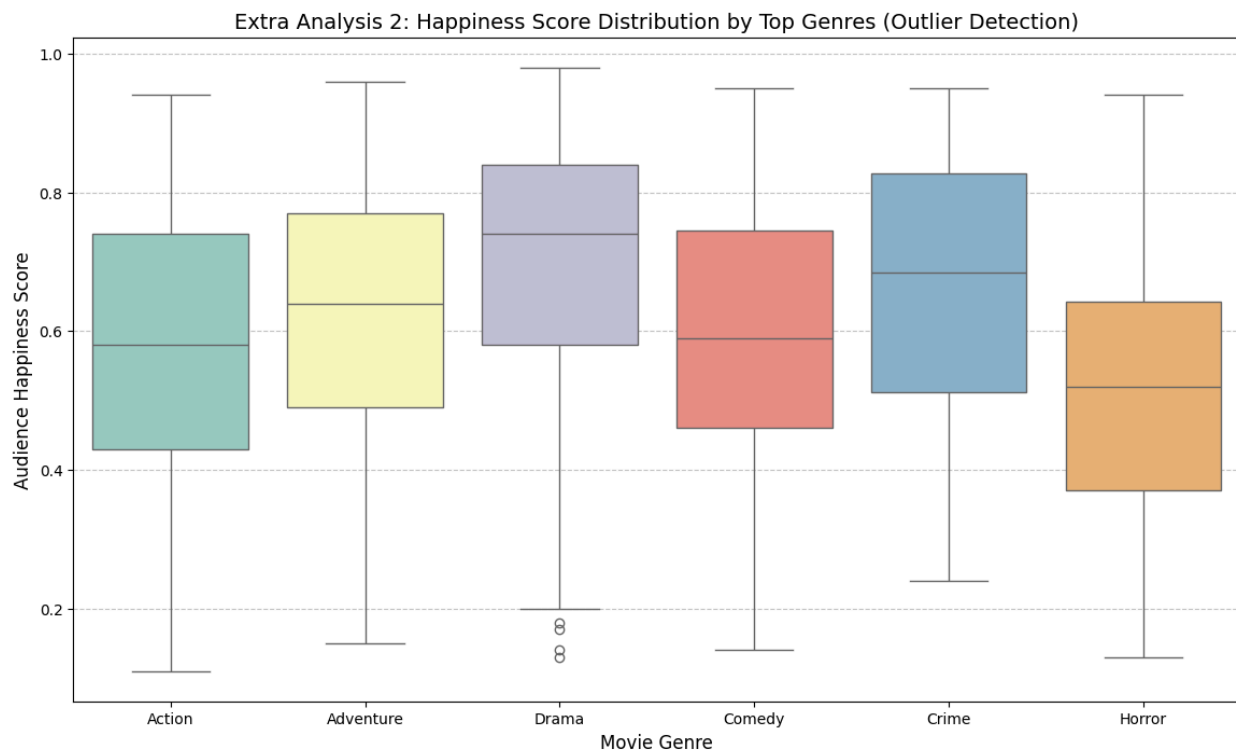


Figure 1: Distribution of Happiness Score across Movie Genres

Phase 2: Unsupervised Learning (K-Means Clustering)

To discover natural, organic archetypes of films in the financial-reception space without forcing predefined historical categories, an unsupervised K-Means Clustering algorithm was applied:

- **Feature Scaling:** Because raw production budgets span hundreds of millions of dollars while engineered happiness scores reside strictly between 0 and 1, input features were transformed using `StandardScaler`. This step is critical; without distance normalization, the massive numerical magnitude of the budget variable would entirely dominate the Euclidean distance matrix, rendering the clustering process mathematically useless.
- **The Elbow Method Optimization:** The optimal hyperparameter K was isolated by calculating the Within-Cluster Sum of Squares (Inertia) across a range of K from 1 to 10. Plotting the results revealed a sharp, clear inflection point precisely at K = 4, defining the ideal cluster structure.

Phase 3: Supervised Learning (Random Forest Regression)

To definitively evaluate whether non-linear machine learning architectures could reliably map financial scales into accurate individual satisfaction scores, a Random Forest Regressor (100 independent decision estimators) was trained. The unified dataset was partitioned using a strict train-test split standard: an 80% Training Set to learn patterns, and a 20% Test Set to evaluate predictive capacity on completely unseen, out-of-sample data. The input feature matrix was defined as $X = [\text{budget}, \text{revenue}]$ and the continuous target vector as $y = [\text{happiness_score}]$.

4. Findings & The Project Paradox

The empirical execution of the data pipeline exposed a profound statistical paradox that highlights the complexity of modeling human emotion with structural metrics, directly addressing the core feedback from the evaluation process:

- **The Correlation Baseline:** The calculated Spearman rank correlation coefficient between production budget and audience happiness yielded a near-zero result: $\text{Rho} = -0.0352$. This strongly indicates the absence of a direct monotonic relationship between financial injection and user appreciation.
- **K-Means Structural Segmentation ($K = 4$):** The unsupervised model cleanly partitioned the industry landscape into four distinct operational quadrants:
 - *Cluster 0 ("The Hidden Gems"):* Lean, low-to-mid budget films that successfully maximized audience satisfaction.
 - *Cluster 1 ("The Underperformers"):* Economical projects that failed to establish a connection with viewers.
 - *Cluster 2 ("The Blockbusters"):* High-budget, high-revenue studio giants grouped together due to extreme financial scale, showing massive, chaotic variance in user ratings.
 - *Cluster 3 ("The Average Performers"):* Standard, predictable mid-budget industry baseline films.

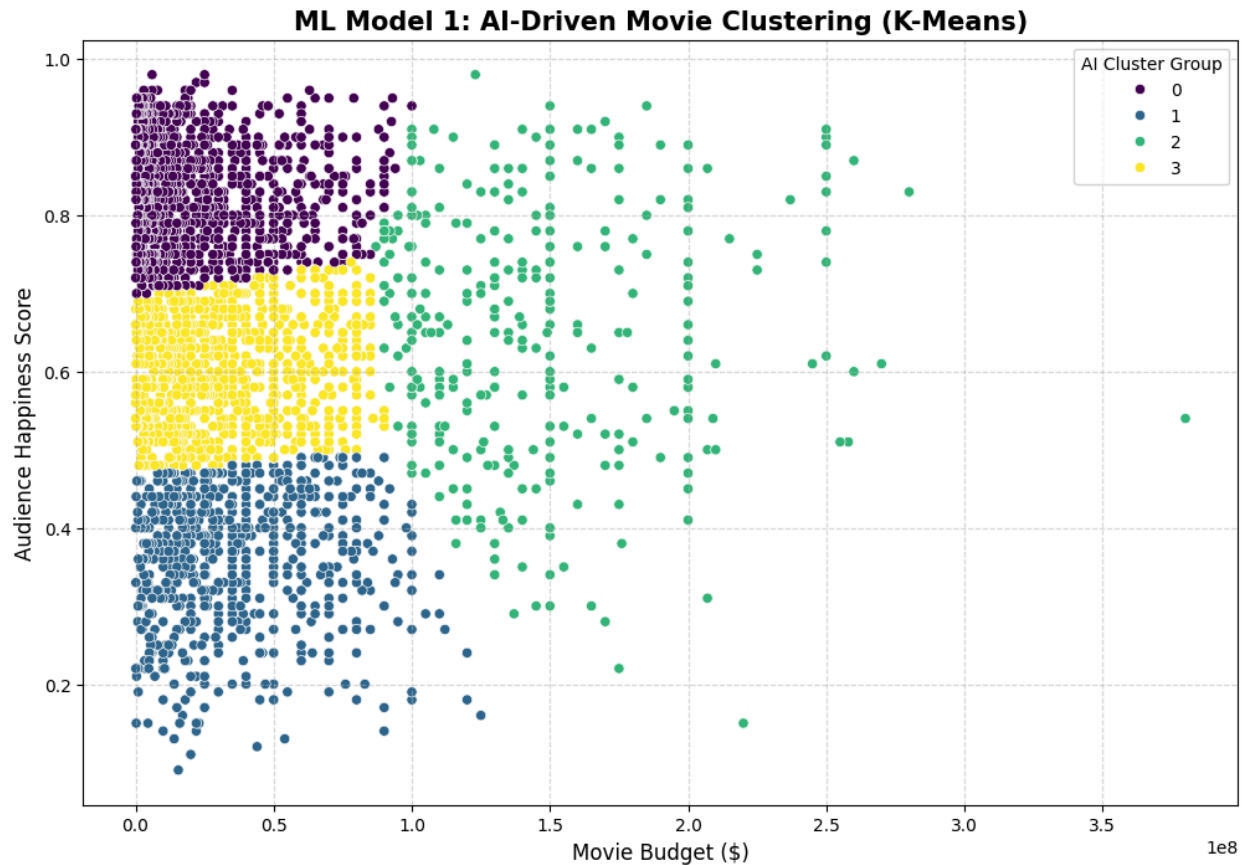


Figure 2: K-Means Clustering Results Mapping Budget vs. Happiness

Supervised Predictive Breakdown: On unseen test data, the optimized Random Forest Regressor returned a high Mean Absolute Error (MAE) of 0.17 and a negative R-Squared score: $R^2 = -0.05$. A negative R^2 indicates that the machine learning model performs worse than a simple baseline horizontal line guessing the mathematical mean of the target variable.

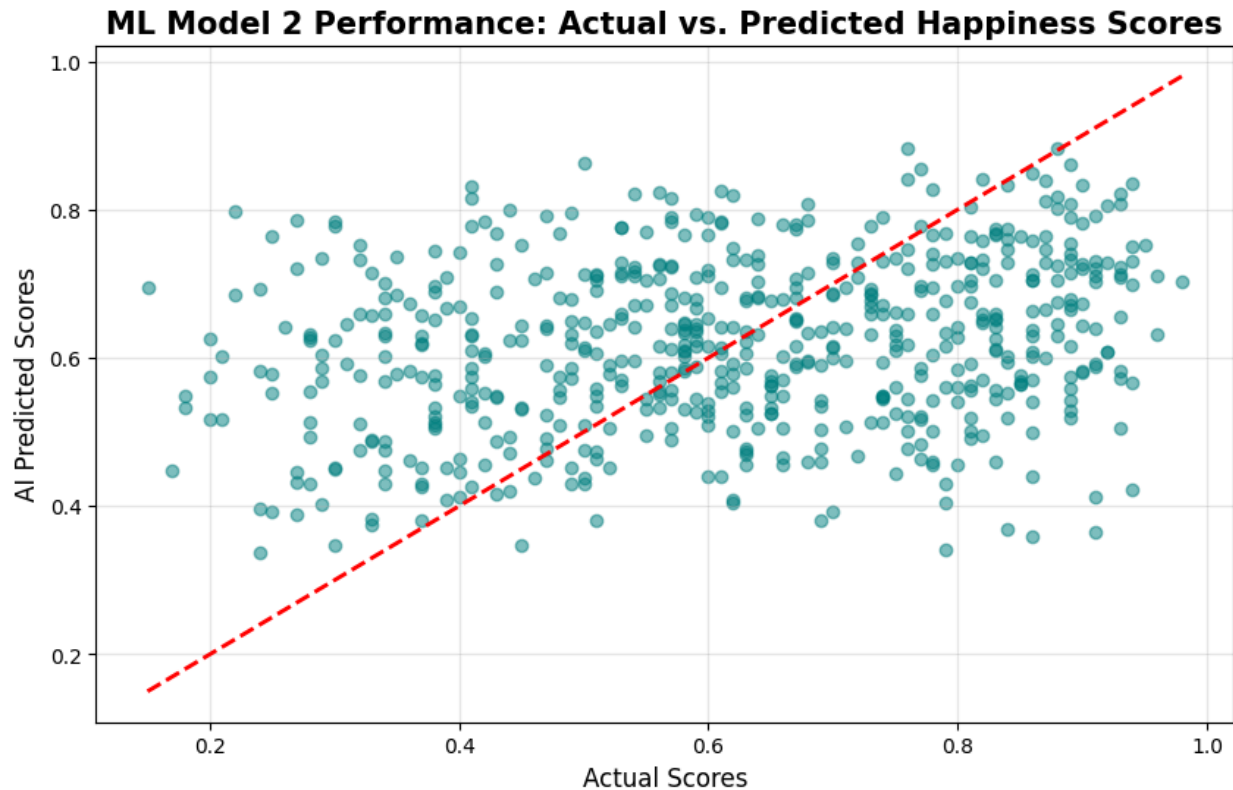


Figure 3: Random Forest Regression Performance (Actual vs. Predicted Scores)

Resolving the Statistical Paradox (Evaluation Feedback Realization)

The defining scientific conclusion of this project lies in explaining why the Kruskal-Wallis test successfully rejected the null hypothesis ($p\text{-value} < 0.05$), indicating a statistically significant relationship, while the predictive machine learning models completely collapsed.

The Kruskal-Wallis test evaluates aggregate group distributions; it mathematically demonstrates that when movies are binned into macro-categories (such as Blockbusters vs. Low Budget indies), their global distribution shapes and operational boundaries are statistically distinct. However, when the Random Forest model attempts to use specific, continuous numerical values of budget and revenue to predict an individual movie's exact `happiness_score`, it hits a wall of localized variance.

A large budget can purchase elite special effects, massive distribution channels, and star power, but it cannot purchase human emotional resonance. Therefore, the continuous "signal" from financial metrics is entirely drowned out by the chaotic "noise" of human sentiment, causing individual predictive models to fail.

5. Limitations & Future Work

Critical Limitations (Omitted Variable Bias):

The primary mathematical limitation causing the supervised regression model to underfit is Omitted Variable Bias. Cinema is fundamentally an art form, and user appreciation is a multi-layered psychological response. The unified dataset utilized in this iteration lacks the true qualitative catalysts of cinematic success, such as script narrative depth, directorial pacing, ensemble cast chemistry, and external cultural timing. Because these variables are unmeasured, financial metrics appear chaotic to the model.

Proposed Future Extensions:

- **Natural Language Processing (NLP):** Integrating an advanced text-processing layer (e.g., Fine-tuned BERT or RoBERTa architectures) to analyze the raw, unstructured textual user and critic reviews from Rotten Tomatoes. This will transform qualitative human words into structured, multi-dimensional emotional embeddings.
- **Categorical Metadata One-Hot Encoding:** Expanding the feature matrix X by explicitly encoding categorical metadata, including studio labels, historical director success coefficients, and primary genre interaction terms.
- **Digital Momentum & Hype Tracking:** Scraping real-time public engagement metrics, such as YouTube trailer view velocities and Twitter/X trend densities, to capture marketing momentum as an independent predictive variable against organic audience reception.